higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 2

13 NOVEMBER 2019

This marking guideline consists of 6 pages.

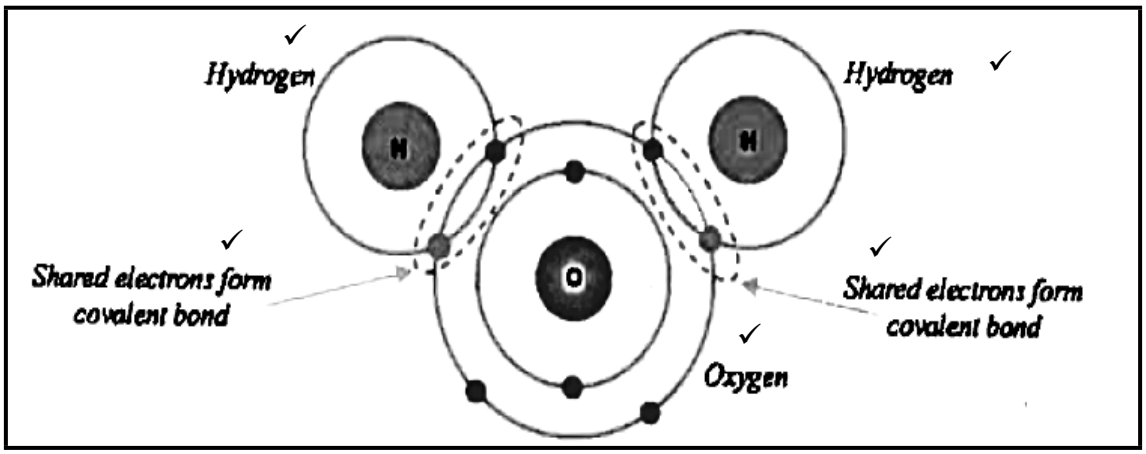
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QUESTION 1

1.1 The electron shells are identified by numbers or letters starting with the innermost shall which is known as the 'K' shell.✓ Note that each electron shell holds different amount of electrons to fill the shell completely.✓ The formula $2n^2$ helps in the determination of the maximum number of electrons present in an orbit,✓ where n is the number of the shell.✓

(4)

1.2



(5)

1.3 1.3.1 Potentiometer
1.3.2 Inductor

(2 × 1) (2)

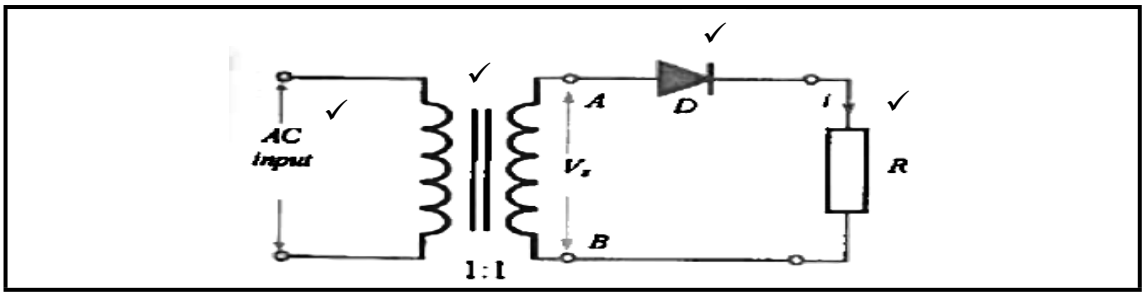
1.4 If a pentavalent atom, such as phosphorous, is added to a pure crystal structure,✓ four of the five valence electrons of the phosphorous will form covalent bonds with its four neighbouring silicon atoms and the fifth valence electron will be left unbound.✓ Since the intrinsic semiconductor has been doped with a controlled amount of pentavalent atoms, many free electrons exist and since the charge on an electron is negative,✓ N-type material is formed.✓ (4)

1.5 Reverse biasing occurs when the external voltage source is connected in such a way that its positive terminal is connected to the N-type material✓ and that the negative terminal is connected to the P-type material.✓ (2)

1.6 Used as a voltage reference
Used in clipping circuits
Used to shift voltages

(accept any use mentioned) (1)

1.7



(4)

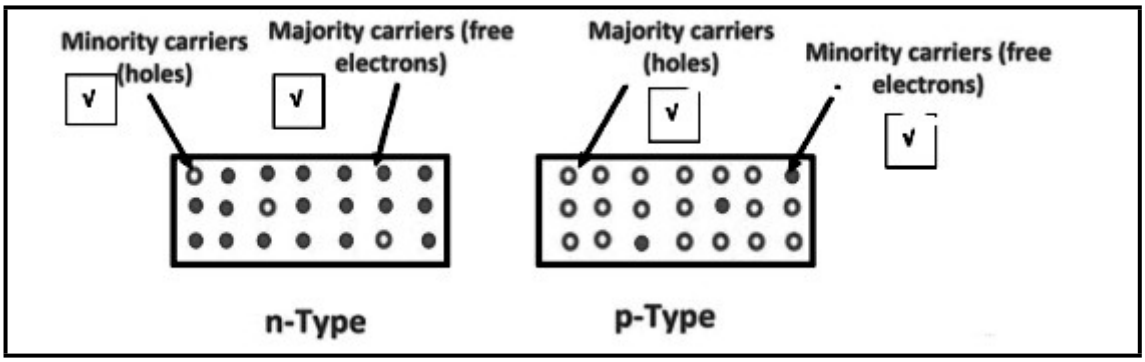
1.8
- Cut-off region
- Active region
- Saturation region

(3)
[25]

QUESTION 2

2.1 Operational amplifier is an IC containing a number of transistors and other components, and amplifies the voltage in a circuit. OR
An operational amplifier is a DC-coupled high-gain electronic voltage amplifier with a differential input and, usually, a single-ended output. (2)

2.2



Alternatively, in words:

In an n-type semiconductor, the electrons are the majority carriers✓ whereas, the holes are the minority carriers✓. In the p-type semiconductor material, the holes are the majority carriers✓, whereas, the electrons are the minority carriers. ✓ (4)

2.3 2.3.1 Wire strippers are used to strip off the insulation from a conducting wire.

2.3.2 Soldering iron is a source of heat that is used to melt solder wire.

2.3.3 Side cutter is used to cut wires and trim leads on the soldered side of a circuit board.

2.3.4 Used to grip small objects, reach awkward places, holding wires, bend loops, and attach wires (accept any use mentioned above)

(4 × 1) (4)

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2.4
- Ignition causing fire or explosion.
- Damage to sensitive electronic components.
- Electric shock to humans, followed by accidents, such as falling.
- Damage to mechanical components. (Any 3 × 1) (3)

2.5
- Whenever handling a circuit board assembly, the operator must be grounded properly.
- Components should be handled by the edges when possible.
- Circuit board assemblies should be handled by the edges.
- Use an ESD safe iron or solder station.
- Use an ESD safe mat to cover your work surface. (5)

2.6 $C_T = C_1 + C_2$
$= 10\mu F + 4.7\mu F$ ✓
$= 14.7\mu F$ ✓
(2)
[20]

QUESTION 3: DIGITAL ELECTRONICS

3.1



3.2

Logic level 0	Logic level 1
Low	High✓
off	On✓
0	1✓
No	Yes ✓

(4)

3.3 3.3.1 Timing diagram shows the time relationship between two or more waveforms with respect to the clock waveform.

3.3.2 Clock signal is a periodic waveform in which intervals between pulses equal the time for one bit.

3.3.3 Positive logic is when a high value of a signal is indicated by a 1 and the low value by a 0 .

(3 × 1) (3)

3.4

$\frac{42}{2} = 21 \quad rem\ 0$

$\frac{21}{2} = 10 \quad rem\ 1$

$\frac{10}{2} = 5 \checkmark \quad rem\ 0 \checkmark$ ✓

$\frac{5}{2} = 2 \quad rem\ 1$

$\frac{2}{2} = 1 \quad rem\ 0$

$\frac{1}{2} = 0 \quad rem\ 1$

$42_{10} = 101010_2 \checkmark$

(4)

3.5

- AND gate
- OR gate
- NAND gate
- NOR gate
- XOR gate
- XNOR gate
- NOT gate

(Any 4 × 1) (4)

3.6 3.6.1 NOR gate (1 × 2) (2)

3.6.2 The output of the NOR gate is high, only when all input voltages are low (2)

3.6.3

INPUT		OUTPUT
A	B	X
0	0	1✓
0	1	0✓
1	0	0✓
1	1	0✓

(4)

3.6.4 Because switch A is damaged, the bulb will only glow when switch B is 0✓ **and**✓ switch A is 0✓ or 1.✓ (4)

3.6.5

INPUT		OUTPUT
A	B	X
0	0	1✓
0	1	0✓
0*	0	1✓
0*	1	0✓

(4)
[35]

QUESTION 4: BASIC COMPUTER COMPONENTS

4.1 4.1.1 C
4.1.2 D

4.1.3 A

(3 × 1) (3)

4.2 4.2.1 VGA (video graphic array) is a port where you can connect your monitor.

4.2.2 RJ-45 port is used to connect a telephone to a computer so that a fax or modem can be used.

(2 × 1) (2)

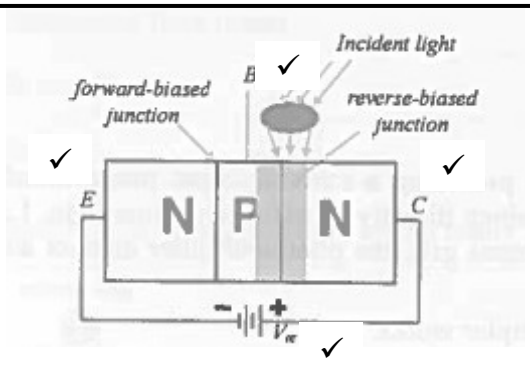
4.3
- Data are stored in a flash memory cell in the form of electrical charge.
- The amount of charge stored depends on the voltage applied to the memory cell, which controls the flow of charge.
- The data contained in the cell depends on whether or not the voltage of the stored charge exceeds a specified threshold voltage.
- The level of the charge passing through the cell is monitored and if the flow of charge is greater than 50% of the charge, it has a value of one.
- If the charge passing through drops below the 50% threshold, the value changes to zero. (5)

[10]

QUESTION 5: TRANSDUCERS USED IN PROCESS CONTROL

5.1
- Flow
- Light
- Pressure
- Displacement or distance
- Liquid level
- Gas flow
- Noise levels
- Thickness
- Force (Any 4 × 1) (4)

5.2



(4)

5.3
- Inaccurate or incorrect readings could easily be taken from a scale.
- The wear and tear of the instrument mechanisms could result in incorrect or inaccurate readings being taken.
- Some instruments are affected by changes in environmental conditions, such as changes in temperature, humidity and electrostatic or magnetic effects. (Any 2 × 1) (2)

[10]

TOTAL: 100

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